Practice Assignment 7 - Not Graded

This assignment will not be graded and is only for practice.
Note: Suppose W_1 and W_2 are subspaces of a vector space V. Then sum of two subspaces is defined as:

$$W_1 + W_2 = \{w_1 + w_2 \mid w_1 \in W_1, w_2 \in W_2\}.$$

One can verify that $W_1 + W_2$ is a subspace of V.

Multiple Select Questions (MSQ):

1 point
Let $V_1 = \mathbb{R}^3$ be a vector space, define a function $\|\cdot\|_i : V_1 \to \mathbb{R}$ for $i = 0, 1, 2, 3$ as:

1. $\|(x, y, z)\|_0 = x + y + z$
2. $\|(x, y, z)\|_1 = |x| + |y| + |z|$
3. $\|(x, y, z)\|_2 = \sqrt{x^2 + y^2 + z^2}$
4. $\|(x, y, z)\|_3 = \max\{|x|, |y|, |z|\}$

and let $V_2 = M_{2\times 2}(\mathbb{R})$ be a vector space, define a function $\|\cdot\|_i : V_2 \to \mathbb{R}$ for $i = 4, 5$ as:

1. $\|A\|_4 = \max\{|a_{11}| + |a_{12}|, |a_{21}| + |a_{22}|\}$, where $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$.
2. $\|A\|_5 = \max\{|a_{11}| + |a_{21}|, |a_{12}| + |a_{22}|\}$, where $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$.

Which of the following option(s) is (are) true.

- [ ] $\|\|_0$ is a norm on V_1.
- [ ] $\|\|_1$ is not a norm on V_1.
- [ ] $\|\|_2$ is a norm on V_1.
- [ ] $\|\|_3$ is not a norm on V_1.
- [ ] $\|\|_4$ is not a norm on V_2.
- [ ] $\|\|_5$ is a norm on V_2

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\|\|_2$ is a norm on V_1.
$\|\|_5$ is a norm on V_2

1 point

Let $\langle ., . \rangle$ denote the standard inner product on R^3, i.e., $\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle = x_1 y_1 + x_2 y_2 + x_3 y_3$. Consider a linear transformation $T : R^3 \rightarrow R$ defined as:

$$T(u) = \langle u, v \rangle,$$

where $v \in R^3$. Which of the following options is (are) true for T?

- [ ] T is one-one for all $v \neq 0 \in R^3$.
- [ ] T is onto for all $v \neq 0 \in R^3$.
- [ ] T is not one-one for every $v \in R^3$.
- [ ] There exists a $v \in R^3$ such that T is an isomorphisim.

No, the answer is incorrect.
Score: 0
Accepted Answers:
T is onto for all $v \neq 0 \in R^3$.
T is not one-one for every $v \in R^3$.

1 point

Which of the following options are true?

- [ ] $V = R^2$ and a function defined as:

$$\langle ., . \rangle : V \times V \rightarrow R$$
$$\langle (x_1, x_2), (y_1, y_2) \rangle = \sum_{i,j=1}^{2} x_i y_j .$$

is not an inner product on the vector space V.

- [ ] $V = M_{2\times 2}(R)$ and $\langle ., . \rangle_1$ and $\langle ., . \rangle_2$ are two inner products on the vector space V, a function defined as: $\langle ., . \rangle = \langle ., . \rangle_1 + \langle ., . \rangle_2$ is an inner product on the vector space V.
- [ ] $V = M_{3\times 3}(R)$ and $\langle ., . \rangle_1$ and $\langle ., . \rangle_2$ are two inner products on the vector space V, a function defined as: $\langle ., . \rangle = \langle ., . \rangle_1 - \langle ., . \rangle_2$ is an inner product on the vector space V.
- [ ] $V = R^2$ and a function defined as:

$$\langle ., . \rangle : V \times V \rightarrow R$$
$$\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_2 - x_2 y_1 .$$

is an inner product on the vector space V.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$V = \mathbb{R}^2$ and a function defined as:

$$\langle .,. \rangle : V \times V \to \mathbb{R}$$
$$\langle (x_1, x_2), (y_1, y_2) \rangle = \sum_{i,j=1}^{2} x_i y_j .$$

is not an inner product on the vector space V.
$V = M_{2\times 2}(\mathbb{R})$ and $\langle .,. \rangle_1$ and $\langle .,. \rangle_2$ are two inner products on the vector space V, a function defined as: $\langle .,. \rangle = \langle .,. \rangle_1 + \langle .,. \rangle_2$ is an inner product on the vector space V.

1 point
Consider a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ defined as $T(x, y, z) = (x + 2y, x - y + cz, 2x + y + dz)$. Let A be the matrix representation of T with respect to the standard ordered bases of $\mathbb{R}^3$ and $B = \begin{bmatrix} 0 & 3 & 2 \\ -1 & -3 & -3 \\ 2 & 3 & 4 \end{bmatrix}$.

Based on the above information which of the followings option(s) is(are) true?

- [ ] If A is similar to B, then rank of the linear transformation T is 2.
- [ ] If A is similar to B, then nullity of the linear transformation T is 2.
- [ ] If $c \neq d$, then A cannot be similar to B.
- [ ] If A is similar to B and $c = 1$, then a basis of the kernel is $\left\{ \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} \right\}$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If A is similar to B, then rank of the linear transformation T is 2.
If $c \neq d$, then A cannot be similar to B.
If A is similar to B and $c = 1$, then a basis of the kernel is $\left\{ \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} \right\}$.

1 point
Consider a vector $v = (1, 2)$ from the inner product space $\mathbb{R}^2$, where inner product is the dot product. Which of the following options is/are true?

- [ ] There exists a vector $u\ (\neq v)$ in $\mathbb{R}^2$ such that $||u|| = ||v||$.
- [ ] There are infinitely many vectors in $\mathbb{R}^2$ such that norms of the vectors are equal to norm of v.

- [ ] There exists a vector u in R^2 such that $\langle u, v\rangle = 0$.
- [ ] There are infinitely many vectors in R^2 such that angle between the vector and v is equal to 90°.

No, the answer is incorrect.
Score: 0
Accepted Answers:
There exists a vector $u\ (\neq v)$ in R^2 such that $||u|| = ||v||$.
There are infinitely many vectors in R^2 such that norms of the vectors are equal to norm of v.
There exists a vector u in R^2 such that $\langle u, v\rangle = 0$.
There are infinitely many vectors in R^2 such that angle between the vector and v is equal to 90°.

1 point
Let L and L' be affine subspaces of R^3, where $L = U$ and $L' = (2,0,1) + U'$, for some vector subspaces U and U' of R^3. Let a basis for U be given by $\{(2,0,1),(1,1,0),(0,1,0)\}$ and a basis for U' be given by $\{(1,0,1),(0,1,1)\}$. Suppose there is a linear transformation $T : U \to U'$ such that $(0,1,0) \in ker(T)$, $T(2,0,1) = (0,1,1)$ and $T(1,1,0) = (1,0,1)$. An affine mapping $f : L \to L'$ is obtained by defining $f(u) = (2,0,1) + T(u)$, for all $u \in U$. Which of the following options are true?

- [ ] $L = \{(x,y,z) \mid x - y - 2z = 0\}$
- [ ] $L = R^3$
- [ ] $L' = \{(x,y,z) \mid x + y - z = 1\}$
- [ ] $L' = R^3$
- [ ] $f(x,y,z) = (x - 2z + 2, z, x - z + 1)$
- [ ] $f(x,y,z) = (x - 2z + 2, \frac{x}{2}, x - z + 1)$
- [ ] $f(x,y,z) = (x - 2z, z, x - z)$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$L = R^3$
$L' = \{(x,y,z) \mid x + y - z = 1\}$
$f(x,y,z) = (x - 2z + 2, z, x - z + 1)$

1 point
Let $L_1 = \{(x,y,z) \mid x + y + 2z = 6\}$ and $L_2 = \{(x,y,z) \mid 2x + y + 2z = 9\}$ be affine subspaces of R^3 and W_1 and W_2 be the corresponding vector subspaces of the given affine subspaces of L_1 and L_2 respectively. Choose the set of correct options

- [ ] $W_1 = \{(x,y,z) \mid x + y + z = 0\}$

- [ ] $W_2 = \{(x, y, z) \mid 2x + y + 2z = 0\}$
- [ ] $W_1 = \{(x, y, z) \mid x + y + 2z = 0\}$
- [ ] $W_2 = \{(x, y, z) \mid 2x + y + 2z = 1\}$
- [ ] $W_1 = \{(x, y, z) \mid x + y + 2z = 1\}$
- [ ] $W_1 \cap W_2 = \{(x, y, z) \mid x = 0, y + 2z = 0\}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$W_2 = \{(x, y, z) \mid 2x + y + 2z = 0\}$
$W_1 = \{(x, y, z) \mid x + y + 2z = 0\}$
$W_1 \cap W_2 = \{(x, y, z) \mid x = 0, y + 2z = 0\}$

Numerical Answer Type (NAT):

Let θ be the angle between the vectors $u = (2, 1, -3)$ and $v = (5, 8, 4)$ and $||.||$ denote the usual length in R^3 with respect to the dot product, then what is $||u||||v||cos(\theta)$?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 6
1 point

Let $\langle ., . \rangle$ denote the standard inner product on R^3, i.e., $\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle = x_1 y_1 + x_2 y_2 + x_3 y_3$. Let v_1 =(1,0,-1), $v_2 = (a, b, c) \in R^3$. Suppose the length of the vector v_2 is 3 and $\langle v_2, (1, -1, 1) \rangle = -3$, $\langle v_2, (-1, 2, 1) \rangle = 1$. If the angle between v_1 and v_2 is 45°, then find the value of $a + b + c$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point

Let $A_{4\times4}$, $B_{4\times4}$ and $C_{4\times4}$ be matrices such that A and B are equivalent matrices and B and C are similar matrices. If rank of the matrix A is 3 then find the determinant of the matrix C.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point